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College Student Binge Eating: Attachment, Psychological Needs Satisfaction, and Emotion Regulation

Suejung Han Soonhee Lee

In this study we examine the mediating roles of psychological needs satisfaction (i.e., autonomy, competence, and relatedness) and emotion regulation difficulties in the relationship between attachment insecurity (i.e., attachment anxiety and avoidance) and binge eating behavior in college students. A total sample of 820 college students participated in this study. Structural equation modeling was performed to test a mediation hypothesis. The mediation model that included psychological needs satisfaction and emotion regulation difficulties as sequential mediators for the link between attachment insecurity and binge eating behavior fit the data well.

Binge eating, consuming a very large amount of food within a short period of time with no sense of control over one's eating (American Psychiatric Association [APA], 2013), is highly prevalent among college students. Researchers have reported that approximately 32.0% to 44.0% college women engage in binge-eating episodes (Napolitano & Himes, 2011; Saules et al., 2009). College transition can be a trigger for disordered eating behaviors like binge eating because students with certain risk factors (e.g., emotional eating tendency) may rely on binge eating to address stress from separation from family or from adjustment to college (Barker & Galambos, 2007). Moreover, a college health survey showed 44.0% of college women and 27.0% of college men were on a diet to lose weight (American College

Health Association, 2010), which has been known as a common trigger for binge eating (e.g., Polivy & Herman, 1985). In addition, according to the annual report of the Center for Collegiate Mental Health (2017), clinicians reported that 13.5% of students who visited college counseling centers in 2015–2016 dealt with disordered eating/body image issues regardless of their primary concerns and 1.9% of students dealt with eating/body image issues as primary concerns. This implies potential high comorbidity of disordered eating behavior with other mental health issues among college students. Moreover, eating disorders increased from 23.0% to 32.0% among females and from 7.9% to 25.0% among male college students (White, Reynolds-Malear, & Cordero, 2011) over the past decade. Therefore, student affair offices, college health services, and college counseling centers have increased their efforts to raise awareness of eating disorders through outreach programs or hired staff practitioners who have a specialty in eating disorders (National Eating Disorder Association, 2013).

Binge eating is a serious physical and mental health threat to college students. It is a precursor to or defining symptom of binge eating disorder and bulimia nervosa (APA, 2013; Dingemans, Bruna, & van Furth, 2002) and associated with health conditions such as obesity and hypertension and mental health conditions such as depression (Darby, Hay, Mond, Rodgers, & Owen, 2007; Saules et al., 2009). Scholars have called for research

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on binge eating at a behavioral level (e.g., Brownstone et al., 2013) rather than as eating disorder categories. This is because (a) disordered eating behaviors manifest themselves heterogeneously even in people with the same diagnosis, (b) treatment interventions are targeted for specific eating behaviors, not disorders, and (c) subclinical levels of binge eating behavior that do not meet the full criteria for eating disorder diagnosis are highly prevalent and still detrimental to physical and psychological functioning (Hoyt & Ross, 2003).

Theoretical Background

According to emotion regulation models of binge eating, people may engage in binge eating when experiencing painful negative self-awareness or emotions (e.g., shame, self-loathing), as avoidance or emotion-focused coping to temporarily avoid them with tasty foods (e.g., Haedt-Matt & Keel, 2011; Heatherton & Baumeister, 1991; Hilbert & Tuschen-Caffier, 2007). A series of research supported this model, demonstrating emotion regulation difficulties and emotional eating tendencies in people who binge eat (e.g., Klein, Skinner, & Hawley, 2013; Mitchell & Mazzeo, 2004). It is still not clearly understood where these emotion regulation difficulties come from and how they contribute to binge eating. In this study, we applied attachment theory and self-determination theory to binge eating to understand its mechanism. Specifically, we examined the mediating roles of basic psychological needs satisfaction and emotion regulation difficulties in the association between attachment insecurity and binge eating.

Attachment Insecurity

Attachment (Bowlby, 1969/1982) refers to an enduring emotional bond with a primary attachment figure (e.g., caregiver, romantic partner). A biologically prewired attachment

system activates when one perceives a threat in the environment. The activated attachment system motivates the person to seek proximity to the attachment figure for safety, comfort, and guidance. Consistent responsiveness from the attachment figure to the proximity seeking is internalized in the form of internal working models of the self (e.g., “I am worthy of consistent care”) and others (e.g., “others are trustworthy and care about me”). Individuals carry internal working models through adulthood that become the framework for developing a sense of autonomy, effectance, and belongingness. Scholars of adult attachment theory (Fraley, Waller, & Brennan, 2000; Hazan & Shaver, 1987) have suggested that adult attachment insecurity might be best represented by two dimensions: attachment anxiety and attachment avoidance. Within this framework, attachment security is operationally defined as low levels of both dimensions. Attachment anxiety represents anxiety about separation and abandonment and perception of an attachment figure’s inconsistent availability and responsiveness. Attachment avoidance represents discomfort with and avoidance of proximity seeking to the attachment figure and perception of the attachment figure’s consistent unavailability or unresponsiveness (Fraley & Shaver, 2000). Insecure attachment has been linked to disturbance in psychological well-being and ineffective coping (for a review, see Mikulincer & Shaver, 2007).

One characteristic of insecure attachment, by definition, is unfulfilled felt security. This implies that individuals with insecure attachment may experience heightened needs for safe haven (i.e., feeling soothed and protected) and secure base (i.e., seeking proximity when upset and seeking exploration when safe; Bowlby, 1969/1982). To regulate these needs, individuals with attachment anxiety may activate the attachment system excessively (i.e., hyperactivating strategy),

become hypersensitive to attachment concerns, and intensify emotional distress. On the other hand, individuals with attachment avoidance may suppress the attachment system (i.e., deactivating strategy), and therefore are suppressive of attachment needs and emotional distress (Mikulincer & Shaver, 2007). However, both hyperactivating and deactivating strategies fail to regulate their concerns effectively and their needs for felt security continue to be unmet.

The relationship between food and those who binge eat may resemble this attachment insecurity. Culturally, offering food may symbolize offering love and affection (Abdussalam, Foster, & Kaferstein, 1989). Food, unlike dismissive or inconsistently unavailable attachment figures, is never rejecting but easily accessible for emotional soothing and comfort (Schlundt & Johnson, 1990). People with anxious attachment may seek proximity to food as a way to meet their emotional needs and soothe themselves under emotional distress. People with avoidant attachment may seek food as a way to find safe haven/secure base without having to acknowledge their suppressed attachment needs or risking anticipated rejection from the actual attachment figure. Or, it is also possible that binge eating may not be the choice of their coping because they may avoid foods in the same way as they avoid proximity seeking to the attachment figure.

Past research supported the association between attachment anxiety and binge eating consistently, but showed mixed findings for the association between attachment avoidance and binge eating (Eggert, Levendosky, & Klump, 2007; Han & Pistole, 2014; Kiang & Harter, 2006). Inconsistent research findings for attachment avoidance seem to parallel to two opposite scenarios that were described above (Larsen, van Strien, Eisinga, Herman, & Engels, 2007). Furthermore, it is possible that the

inconsistent association between attachment avoidance and binge eating was due to limited understanding of the mechanism of the link. The current study proposes psychological needs satisfaction and emotion regulation difficulties as potential mediators that may elucidate the complex relationship between attachment insecurity and binge eating. As a theoretical perspective on psychological needs, we draw our rationale from self-determination theory (Ryan & Deci, 2000).

Basic Psychological Needs

Self-determination theory conceptualizes that human beings have three basic psychological needs for autonomy, competence, and relatedness that contribute to health, integrity, and psychological well-being (Ryan & Deci, 2000). The need for autonomy refers to the need to feel volitional and to experience a sense of choice in initiating one's action. The need for competence refers to the need to feel effective in attaining desired outcomes. The need for relatedness refers to the need to feel connected to others and cared for. Considering that attachment security develops when significant others are promptly and consistently responsive to individuals' needs, the associations between attachment and psychological needs seem obvious. People with attachment insecurity may have developed chronic sense of unfulfilled (or heightened desire for fulfilling) psychological needs because their (hyperactivated or deactivated) attachment system repeatedly fails to offer felt security (Bowlby, 1969/1982; Mikulincer & Shaver, 2007). They may feel that they have no choice but to desperately seek for proximity or avoid proximity (i.e., low autonomy), are not effective in obtaining what they want because they seek support excessively or try to rely on themselves compulsively (i.e., low competency), and do not have a strong emotional bond with their attachment

figures due to unpredictable responses or consistent rejection (i.e., low relatedness). Among the few studies conducted on the link between adult attachment and basic psychological needs, Wei, Shaffer, Young, and Zakalik (2005) reported that both attachment anxiety and attachment avoidance are negatively associated with psychological needs satisfaction in undergraduate students. Felton and Jowett (2013) also reported negative association between insecure attachment and psychological needs satisfaction in the context of parent-athlete and coach-athlete relationships. In addition, La Guardia, Ryan, Couchman, and Deci (2000) showed that within-person variations in attachment security coincided with gratification of individuals' basic psychological needs by significant others such as parents, romantic partner, and friends.

Previous research indicated that needs satisfaction is related to well-being such as vitality and physical health (e.g., Ryan & Deci, 2008) and positive and negative affect (e.g., Ryan, Bernstein, & Brown, 2010). Self-determination theory states that, when people chronically fail to fulfill basic psychological needs, they may engage in rigid or dysfunctional behavior (e.g., drinking, smoking) to compensate for those unmet needs. Disordered eating behaviors such as binge eating can be understood as such compensatory behaviors in response to chronically unfulfilled psychological needs (Verstuyf, Patrick, Vansteenkiste, & Teixeira, 2012). Past studies have found that unfulfilled psychological needs were associated with binge eating (Schüler & Kuster, 2011; Verstuyf, Vansteenkiste, & Soenens, 2012). A daily diary study further demonstrated that fluctuations in needs frustration covaried with fluctuations in binge eating symptoms at the intrapersonal level (Verstuyf, Vansteenkiste, Soenens, Boone, & Mouratidis, 2013).

Emotion Regulation Difficulty

Difficulty with regulating negative emotions is closely tied to psychological needs because unmet psychological needs would cause negative emotions (Ryan et al., 2010). Emotion regulation refers to "extrinsic and intrinsic processes responsible for monitoring, evaluating, and modifying emotional reactions" (Thompson, 1994, pp. 27–28). It is notable that this definition includes the coregulation of emotion in a relationship context (i.e., attachment) in addition to the own internal emotion regulation process. Developmentally, children in distress internalize how their attachment figure soothes them and develop their own emotion regulation capacity in their internal working model. Therefore, insecure attachment causes limited and less optimal emotion regulation capacity (Fuendeling, 1998).

As mentioned above, the emotion regulation model of binge eating posits that people would engage in binge eating to deal with negative emotions and avoid painful self-awareness (Heatherton & Baumeister, 1991). Research supports that people who binge eat suffer from negative emotions such as boredom, loneliness, anxiety, anger, or depression that precede binge eating, and they are absorbed in the taste of foods to be distracted from such emotions (Wolfe, Baker, Smith, & Kelly-Weeder, 2009). However, such a temporary relief from the negative feelings tends to be quickly replaced by rather sustaining negative self-conscious emotion (e.g., guilt, shame, disgust) and other unsettling emotion (e.g., anxiety, helplessness, irritation, frustration; e.g., Hilbert & Tuschen-Caffier, 2007). The vicious cycles of emotion regulation difficulties through binge eating accentuates the ineffectiveness of binge eating as a coping strategy (e.g., Klein et al., 2013).

Mediation Model

In sum, we propose that insecurely attached individuals have unfulfilled psychological needs and difficulty regulating their negative emotions that may arise from unfulfilled needs, and therefore may rely on binge eating. In other words, both psychological needs satisfaction and emotion regulation difficulties may mediate the association between attachment insecurity and binge eating. Research on mediation models with either psychological needs satisfaction or emotion regulation difficulties is limited. Among the few, Han and Pistole (2014) demonstrated the mediating role of emotion regulation difficulty in the association between both attachment anxiety and attachment avoidance and binge eating in college students. Tasca and his colleagues (2009) showed that emotional reactivity mediated the association between attachment anxiety and eating disorder symptoms, whereas emotional deactivation did not mediate the association between attachment avoidance and eating disorder symptoms. However, their study used aggregated eating disorder symptoms rather than specific binge eating behavior. Other studies tested similar mediation models but with different constructs. Verstuyf, Vansteenkiste, and Soenens (2012) showed the mediating role of diet-specific need frustration in the associations between eating regulation goals and binge eating. Wei et al. (2005) showed the mediating role of psychological needs satisfaction in the association between attachment insecurity and other adjustment outcomes (i.e., shame, depression, and loneliness). To our knowledge, no study has tested the mediating roles of psychological needs satisfaction and emotion regulation difficulties in sequence in the association between attachment insecurity and binge eating.

We examined the interrelationship among

attachment insecurity, psychological needs satisfaction, and emotion regulation difficulties in their contribution to binge eating simultaneously. Specifically, we hypothesized that both psychological needs satisfaction and emotion regulation difficulties would mediate the association between attachment insecurity and binge eating, where psychological needs satisfaction would mediate the association between attachment insecurity and emotion regulation difficulties, and emotion regulation difficulties would mediate the association between psychological needs satisfaction and binge eating (Figure 1). As this is a cross-sectional study, the focus is to shed light on the interconnection among attachment, needs satisfaction, emotion regulation, and binge eating, rather than to examine any chronological or causal sequence of mechanisms that lead to binge eating behavior.

METHOD

Participants

A total of 1,059 college students at a regional US Midwestern university were recruited via e-mail advertisement from the university research office. Respondents were offered an incentive through a drawing for a \$20 gift card (with 1:200 odds) or research participation credits if they were taking a General Psychology class. Among the initial respondents, 218 respondents were removed for failing to complete more than half of the survey, and then 21 respondents were excluded for incomplete items on any of the measures. As the majority of the excluded did not answer demographic questions, attrition analyses on those variables were not conducted. A final sample of 820 participants was used for analyses in the study. They were 253 (30.9%) male and 562 (68.5%) female students with 5 not reporting (0.6%); ages ranged from 17 to 56 years with the mean

age of 21.02 ($SD = 5.12$). The sample was mostly European Americans (743, 90.6%), with 2 African Americans (0.2%), 4 Latino/as (0.5%), 26 Asian Americans (3.2%), 12 multiracial (1.5%), 20 international (2.5%), 10 other (1.2%), and 3 not reporting (0.4%) their race/ethnicity, which represented the racial/ethnic composition of the student population in that university. Missing value analyses with the final sample showed that only 0.56% of the total data was missing and there was no variable with 2.00% or more missing values. As the proportion of the missing data was inconsequential (Schafer, 1999), mean substitutes were used for descriptive analyses and full information maximum likelihood method was used in a structural equation modeling analysis for hypothesis testing.

Instruments

Attachment. The 12-item Experiences in Close Relationship–Short Form (ECR-S; Wei, Russell, Mallinckrodt, & Vogel, 2007) was used to assess adult attachment. It includes two attachment dimensions: attachment anxiety (6 items; e.g., “I need a lot of reassurance that I am loved by my partner”) and attachment avoidance (6 items; e.g., “I am nervous when partners get too close to me”). Participants rated each item on a 7-point Likert-type scale ranging from 1 (*disagree strongly*) to 7 (*agree strongly*). Higher scores indicate higher levels of attachment anxiety and attachment avoidance respectively. Cronbach’s alphas coefficients were reported to range from .78 to .86 for attachment anxiety and from .78 to .88 for attachment avoidance (Wei et al., 2007). In our study, they were .78 for attachment anxiety and .86 for attachment avoidance.

Needs Satisfaction. The 21-item Basic Psychological Needs Satisfaction Scale (BNS-G; Gagné, 2003) was used to measure basic psychological needs fulfillment. The BNS-G includes three basic needs: (a) autonomy

(7 items; e.g., “I generally feel free to express my ideas and opinions”), (b) competence (6 items; e.g., “People I know tell me I am good at what I do”), and (c) relatedness (8 items; e.g., “I really like the people I interact with”). Participants rated each item on a 7-point Likert-type scale, ranging from 1 (*not at all true*) to 7 (*very true*). Higher scores indicate higher levels of needs satisfaction. Cronbach’s alpha coefficients were reported to be .69 for autonomy, .71 for competence, and .86 for relatedness (Gagné, 2003). In this study, they were .70 for autonomy, .73 for competence, and .82 for relatedness.

Emotion Regulation Difficulties. The 36-item Difficulties in Emotion Regulation Scale (DERS; Gratz & Roemer, 2004) was used to measure emotion regulation difficulty. The DERS includes six dimensions: (a) acceptance (6 items), not accepting one’s feeling by having a negative emotional reaction to the feeling (e.g., “When I am upset, I feel guilty for feeling that way”); (b) goals (5 items), difficulty engaging in goal-directed behavior when experiencing negative emotions (e.g., “When I am upset, I have difficulty concentrating”); (c) impulse (6 items), difficulty remaining in control of behavior while experiencing negative emotions (e.g., “When I am upset, I lose control over my behaviors”); (d) awareness (6 items), a lack of emotional awareness (e.g., “When I am upset, I pay attention to my feelings”; reverse scored); (e) strategies (8 items), limited access to emotion regulation strategies (e.g., “When I am upset, I believe that there is nothing I can do to make myself feel better”); and (f) clarity (5 items), lacking emotional clarity when upset (e.g., “I have no idea how I am feeling”). Participants rated items on a 5-point Likert-type scale, ranging from 1 (*almost never*) to 5 (*almost always*). Higher scores reflect more emotion regulation difficulties. The DERS showed construct validity and predictive validity

(Gratz & Roemer, 2004). Cronbach's alphas were reported to be .93 for the total score and range from .80 to .89 for the subscales (Gratz & Roemer, 2004). In our study, they were .93 for the total score and ranged from .84 to .92 for the subscales.

Binge Eating. The 16-item Binge Eating Scale (BES; Gormally, Black, Daston, & Rardin, 1982) was used to measure binge eating behaviors and associated feelings/cognitions (e.g., feeling a lack of control). Each item lists three to four statements that range from normal to problematic eating and each statement was assigned a weight (0–3) consistent with the extent to which it describes binge eating severity. For example, an item has four statements such as (a) “I rarely eat so much food that I feel uncomfortably stuffed afterwards” (weighted 0); (b) “Usually about once a month, I eat such a quantity of food, I end up feeling very stuffed” (weighted 1); (c) “I have regular periods during the month when I eat large amounts of food, either at mealtime or at snacks” (weighted 2); and (d) “I eat so much food that I regularly feel quite uncomfortable after eating and sometimes a bit nauseous” (weighted 3). Scores are summed such that higher scores indicate severer binge eating. The BES demonstrated concurrent validity by differentiating people with severe binge eating from people with mild or no binge eating (Gormally et al., 1982). Cronbach's alpha was reported to be .91 (Han & Pistole, 2014) and was .89 in our study.

RESULTS

Preliminary Analyses

Means, standard deviations, and zero-order correlations are presented in Table 1. Most variables were significantly correlated with one another in an expected direction (r s ranging from .14 to .71), except for the nonsignificant correlations between attachment avoidance and

difficulty with goal directedness and between lack of awareness of emotions and difficulty with goal directness. The correlations were generally modest in magnitude, suggesting acceptable levels of collinearity for the proposed multivariate analyses (Tabachnick & Fidell, 2007). Skewness and kurtosis analyses demonstrated that the BES scores were positively skewed. About 10.5% ($n = 86$) endorsed moderate (i.e., a score between 18 and 26) binge eating behavior, and 3.7% ($n = 30$) severe levels (i.e., a score of 27 and above) of binge eating behavior, while about 86.0% students endorsed none to mild levels of binge eating behavior (i.e., a score of 17 and below; cutoff scores were based on Marcus, Wing, & Hopkins, 1988). A square-root transformation of the BES scores resulted in a normal distribution and was used for the following analyses.

Main Analysis

A structural equation modeling (SEM; Schumacker & Lomax, 2004) analysis was performed using IBM SPSS® Amos™ (version 22.0) to test our hypothesis that attachment insecurity and binge eating would be associated through psychological needs satisfaction and emotion regulation difficulties. SEM allows examination of complex interrelationships among variables simultaneously while controlling for measurement errors and therefore is suitable for testing our model of a serial mediation relationship. The item-to-construct balance method, assigning items to parcels based on the magnitude of the item–construct association, was used to create parcels that represent the latent construct as equally as possible (Little, Cunningham, Shahar, & Widaman, 2002). Three parcels were created as indicators for attachment anxiety and attachment avoidance separately, and four parcels were created for binge eating. For emotion regulation difficulties, the six

DERS subscales were used as indicators. For psychological needs satisfaction, three BNS-G subscales were used as indicators. The sample size ($N = 820$) was more than adequate to detect a small effect size with 5 latent variables with 18 indicators, with the minimum sample size recommended being 463. All latent constructs were allowed to correlate with one another in the measurement model. Four fit indices were used to test the models' goodness of fit: chi-square (nonsignificant results indicate a good fit), comparative fit index (CFI; values of .90 or greater indicate a good fit), root mean square error of approximation (RMSEA; values of .05 or less indicate a good fit), and expected cross-validation index (ECVI; no recommended cutoff value, but lower values indicate a better fit). Full information maximum likelihood method was used in Amos to handle missing data in calculating estimates when missing data is encountered in performing the analysis. Standardized root mean square residual (SRMR) was not used as a fit index, because it cannot be calculated in Amos when data are missing.

Measurement Model. A confirmatory factor analysis (CFA) using maximum likelihood estimation method was conducted to test goodness of fit for the measurement model. The fit indices indicated a marginal

TABLE 1. Means, Standard Deviations, and Correlations

	M	SD	Range	1	2	3	4	5	6	7	8	9	10	11	12
1. ANX	22.08	6.86	6-42	—	.20***	-.29***	-.30***	-.26***	.33***	.36***	.34***	.14***	.45***	.32***	.23***
2. AVD	16.10	6.88	6-42		—	-.35***	-.35***	-.32***	.21***	.07	.16***	.35***	.19***	.34***	.16***
3. Auto	34.58	6.07	7-47			—	.64***	.63***	-.43***	-.30***	-.38***	-.37***	-.51***	-.52***	-.29***
4. Comp	30.76	5.59	6-42				—	.58***	-.40***	-.34***	-.39***	-.36***	-.54***	-.51***	-.30***
5. Relate	45.11	7.21	8-56					—	-.34***	-.20***	-.43***	-.33***	-.50***	-.40***	-.25***
6. Accept	12.67	5.69	6-30						—	.43***	.53***	.24***	.68***	.50***	.32***
7. Goal	14.35	4.80	5-25							—	.46***	.05	.57***	.31***	.21***
8. Impulse	10.51	4.37	6-30								—	.22***	.71***	.45***	.28***
9. Aware	14.72	4.89	6-30									—	.23***	.52***	.16***
10. Strategy	15.97	6.44	8-40										—	.52***	.36***
11. Clarity	10.82	3.83	5-25											—	.30***
12. BE	9.45	7.70	0-46												—

Notes. $N = 820$; 1. ANX = attachment anxiety; 2. AVD = attachment avoidance from Experiences in Close Relationship-Short Form (ECR-S); 3. Auto = autonomy; 4. Comp = competence; 5. Relate = relatedness from Basic Psychological Needs Satisfaction scale (PNS-G); 6. Accept = lack of acceptance of emotions; 7. Goal = difficulty with goal directedness; 8. Impulse = difficulty with impulse control; 9. Aware = lack of awareness of emotions; 10. Strategy = limited emotion regulation strategies; 11. Clarity = lack of clarity of emotions from Difficulties in Emotion Regulation Scale (DERS); 12. BE = binge eating severity from Binge Eating Scale (BES). All correlations were significant at $p < .001$ except nonsignificant correlations between Goal and AVD and between Goal and Aware.

*** $p < .001$.

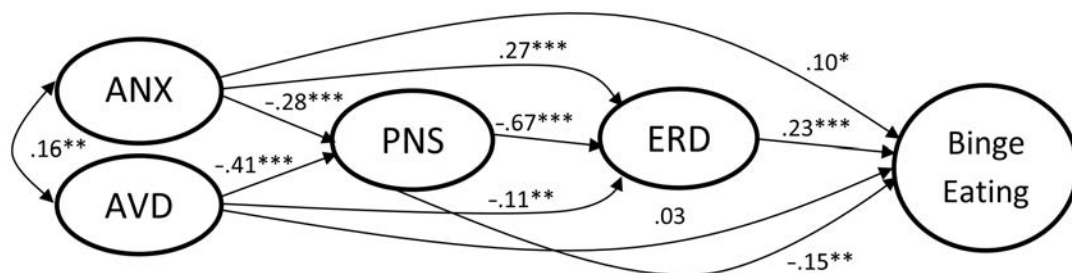


FIGURE 1. SEQUENTIAL MEDIATION MODEL

Note. ANX = attachment anxiety; AVD = attachment avoidance; PNS = psychological needs satisfaction; ERD = emotion regulation difficulties.

* $p < .05$. ** $p < .01$. *** $p < .001$.

fit, $\chi^2(142, n = 820) = 749.95, p < .001$, CFI = .93, RMSEA = .072 (90% confidence interval [.067, .077]), ECVI = 1.08. The DERS awareness path was deleted to improve the model because the path coefficient was below .50 (i.e., .33), following Schumacker and Lomax's (2004) recommendation. This modified measurement model produced an adequate model fit, $\chi^2(125, n = 820) = 475.67, p < .001$, CFI = .97, RMSEA = .053 (90% CI [.059, .064]), ECVI = .74, and was significantly improved from the original model, $\Delta\chi^2(17, n = 820) = 274.28, p < .001$. Therefore, the indicators represented each latent variable sufficiently well in the modified model. All the latent variables were significantly correlated with each other in the expected directions ($p < .001$).

Structural Model and Hypothesis Testing.

To examine our mediation hypothesis, a structural model was tested using the maximum likelihood method. The results showed that the mediation model fit the data adequately, $\chi^2(125, n = 820) = 475.67, p < .001$, CFI = .96, RMSEA = .059 (90% CI [.053, .064]), ECVI = .737. The direct effect from attachment to binge eating was significant for attachment anxiety, $\beta = .10, p = .02$, but not for attachment avoidance, $\beta = .03, p = .55$; all of the other paths were significant in the expected directions,

$\beta = -.28, p < .001$ for attachment anxiety–needs satisfaction, $\beta = -.41, p < .001$ for attachment avoidance–needs satisfaction, $\beta = -.67, p < .001$ for needs satisfaction–emotion regulation difficulties, $\beta = .23, p < .001$ for emotion regulation difficulties–binge eating, $\beta = -.15, p = .03$ for needs satisfaction–binge eating. Asymmetric confidence interval tests using PRODCLIN (MacKinnon, Fritz, Williams, & Lockwood, 2007) revealed that all the mediation effects were significant. Specifically, psychological needs satisfaction significantly mediated the association between attachment and emotion regulation difficulties, 95% CI [.05, .09] for attachment anxiety, 95% CI [.06, .10] for attachment avoidance. Emotion regulation difficulties significantly mediated the association between psychological needs satisfaction and binge eating, 95% CI [-.25, -.07], supporting our hypothesis.

DISCUSSION

The present study examined a possible psychological mechanism of binge eating behavior in the context of attachment theory. Specifically, the results indicated that the link between attachment insecurity and binge eating is accounted for by two mediators, basic psychological needs and emotion regulation

difficulties. In other words, insecure attachment may contribute to binge eating through the mechanisms of lower level of psychological needs satisfaction and emotion regulation difficulties. Previous studies showed that attachment insecurity, psychological needs satisfaction, and emotion regulation difficulties are associated with binge eating behavior independently. The current study examined the interrelationship of these attachment-related process variables simultaneously in predicting binge eating. We demonstrated that psychological needs satisfaction mediated the link between attachment and emotion regulation difficulties and that emotion regulation difficulties mediated the link between psychological needs satisfaction and binge eating behavior.

The current result is in line with well-validated psychological intervention for disordered eating behaviors. For example, attachment is a core component of group psychodynamic-interpersonal therapy (Tasca et al., 2011) and emotion-focused therapy (Dolhanty & Greenberg, 2009; Robinson, Dolhanty, & Greenberg, 2015); emotion regulation is a key component in dialectical behavior therapy (DBT) for eating disorders (Bankoff, Karpel, Forbes, & Pantalone, 2012; Linehan & Chen, 2005). In addition, our study uniquely expands the attachment and emotion regulation models by identifying basic psychological needs as a separate but interconnected component in the mechanism of binge eating.

Furthermore, our model can be placed in the larger context of college student development. College students undergo development through accomplishing emerging adulthood tasks such as developing competence, managing emotions, moving through autonomy toward interdependence, and developing mature interpersonal relationships (Chickering & Reisser, 1993). Our model indicates how impediment in this normative development process would contribute to

unhealthy behavior such as binge eating. Enhancing psychological needs fulfillment and emotion regulation would not only promote college student development in general but also intervene with and reduce the path to binge eating specifically.

Practical Implications

The proposed mediation model highlights intervention foci for students with binge eating behavior. First, student affairs professionals or college counselors may consider attachment insecurity when designing interventions or preventive programs for disordered eating. Campus wellness or health promotion offices may provide informational resources (e.g., poster, brochure, websites) that explain how transition to college can be an attachment disruption and trigger for disordered eating (Barker & Galambos, 2007). They also could add an attachment-related information session to their existing campus-wide events (e.g., new student orientation, body image workshops, National Eating Disorders Awareness Week). Providing workshop on interpersonal effectiveness to reduce attachment anxiety for students in difficult relationship and increase attachment security for students creating new connections on campus would be an example. Educating residence hall assistants and peer advisors about their potential attachment roles for students (e.g., safe haven for soothing distress from academic stress) could be useful. College counselors also could help students gain insight in potential attachment meaning of their binge eating (e.g., proximity seeking to attachment figures), and serve as an attachment figure on campus to enhance their security (Keating et al., 2014).

Second, student affairs professionals may provide educational resources or workshops on the importance of three basic psychological needs and life domains where their basic psychological needs could be fulfilled. In addition, college counselors may help students

become more aware of what they actually need when they feel compelled to binge eat and collaboratively examine where their autonomy (e.g., separation and individuation from parents in career choices), competence (e.g., class performance lower than own expectations), and relatedness (e.g., lack of social network on campus, separation from family) needs are unfulfilled. Health promotion professionals may consider providing education on functional and effective strategies of fulfilling psychological needs (e.g., setting healthy interpersonal boundaries for autonomy, making small goals for success experiences for competence, joining clubs that fit their values and interests for relatedness) and their connection to psychological well-being as a way to intervene with and prevent binge eating.

Last, improving emotion regulation capacity can be another entry to intervene with and prevent binge eating. Campus wellness programs or therapy groups targeting effective emotion regulation (e.g., DBT–emotional regulation skills group; Linehan & Chen, 2005) or making such information readily accessible (e.g., posters around campus, brochures in student affairs offices) may also be useful for not only enhancing general emotional health but also preventing dysfunctional behavior such as binge eating. In therapy, college counselors may help students learn to accept emotion nonjudgmentally and develop and practice functional emotion regulation strategies (Robinson et al., 2015).

Limitations

This study has limitations that warrant future research. First, the mediation model was tested on cross-sectional data and correlational by nature. Longitudinal studies or daily diary studies may answer the causal sequences of binge eating behavior. Second, there is room for using alternative measures to assess psychological needs or emotion regulation difficulties.

Self-determination theory researchers have proposed the concept of needs thwarting to explain disordered behaviors (e.g., Sheldon, Abad, & Hinsch, 2011). Verstuyf et al. (2013) showed that psychological needs satisfaction was unrelated to binge eating, whereas needs thwarting was significantly related to binge eating. In addition, Tasca et al. (2009) used the Differentiation of Self Inventory (DSI-R; Skowron & Friedlander, 1998) to assess emotional reactivity and emotional deactivation as emotion regulation difficulties for attachment anxiety and attachment avoidance respectively. Future studies may test the proposed model utilizing such alternative measures. Third, we did not have enough power to conduct gender comparisons (minimum needed $n = 463$ for men). Even though gender is neither the main focus of this study nor a differentiating factor in attachment theory, future studies may examine whether gender moderates the proposed mediation links. Finally, we used a convenient sample of nonclinical college students who are mostly Whites. Future research should examine the applicability of the proposed mediation model in other racial/ethnic groups or clinical populations.

In conclusion, our study highlights the two attachment process variables—psychological needs and emotion regulation—as the mechanism of binge eating for individuals with attachment insecurity and potential foci for intervening with or reducing risk for college students' binge eating. Our results suggest the importance of higher education professionals developing and providing interventions and preventive programs for disordered eating that focus on boosting attachment security, psychological needs satisfaction, and effective emotion regulation.

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